



SENIOR SCHOOL · COMPETENCY BASED CURRICULUM

GRADE 10

PHYSICS

Study Pamphlet & Revision Notes

Covering all four strands of the KICD Grade 10 Physics Curriculum:

**Mechanics & Thermal Physics · Waves & Optics ·
Electricity & Magnetism · Environmental & Space Physics**

Prepared for learners aged 14–16 years

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STRAND 1: Mechanics and Thermal Physics

How forces, pressure, heat and machines behave and interact in the physical world

1.1 Introduction to Physics

Physics is the branch of science that studies matter, energy, and the forces that act between them, and how these explain the behaviour of the physical world — from the smallest atom to the largest galaxy.

Branches of Physics

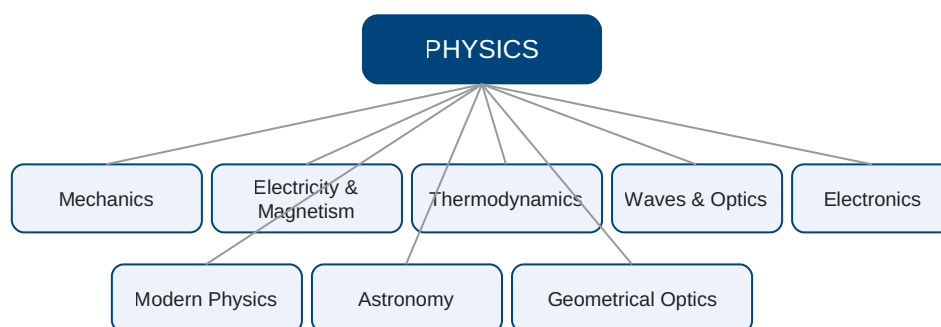


Fig 1.1: The main branches of Physics as a field of study

Why Physics Matters in Everyday Life

- Explains natural events: why the sky is blue, why a boat floats, why a matatu needs brakes to stop
- Powers technology: mobile phones, solar panels, motorbikes (boda boda), electricity generation at Gitaru and Olkaria
- Improves farming: irrigation pumps, greenhouse technology, solar dryers
- Supports medicine: X-rays, ultrasound scanners, radiotherapy

Physics and Other Fields of Study

Physics links closely with **Mathematics** (equations, graphs), **Chemistry** (atomic structure, energy changes), **Biology** (optics of the eye, forces in the body), **Geography** (climate, weather), and **Engineering** (structures, machines).

Careers Related to Physics

Engineer

Pilot

Astronomer

Medical Physicist

Electrician

Meteorologist

Data Scientist

Solar Technician

Radiographer

Research Scientist

Quick Revision: Physics = study of matter, energy & forces. It has many branches, connects to other subjects, and opens up many STEM careers.

1.2 Pressure

Pressure is the force acting normally (perpendicularly) per unit area.

$$\text{Pressure (P)} = \text{Force (F)} \div \text{Area (A)} \rightarrow P = F/A \quad (\text{Units: Pascals, Pa} = \text{N/m}^2)$$

Atmospheric Pressure

The air around the Earth (the atmosphere) has weight, and this weight exerts pressure on everything below it — this is **atmospheric pressure**. At sea level, it is about 101,325 Pa (1 atmosphere).

Everyday evidence: A drinking straw works because sucking reduces air pressure inside the straw, so the greater atmospheric pressure outside pushes the drink up. Similarly, tightly-sealed tins "pop" when opened because of the pressure difference.

Pressure in Fluids

Pressure in a fluid (liquid or gas) increases with depth and depends on the density of the fluid.

$$P = \rho gh$$

where ρ = density of fluid (kg/m^3), g = acceleration due to gravity (10 m/s^2 or 9.8 m/s^2), h = depth below the surface (m)

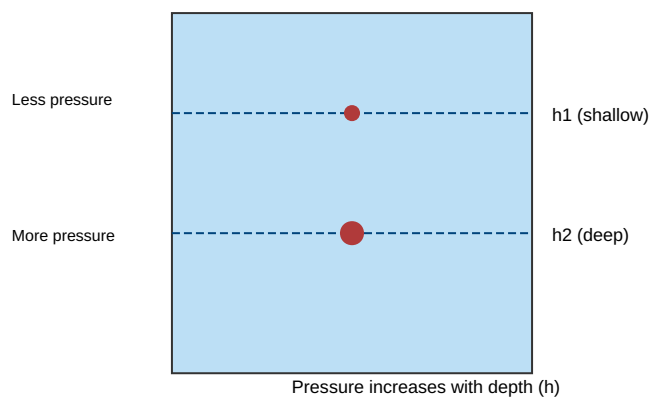


Fig 1.2: Pressure at different depths in a fluid — the deeper point (h2) has greater pressure

Transmission of Pressure in Fluids (Pascal's Principle)

Pressure applied to an enclosed fluid is transmitted equally in all directions. This principle is used in **hydraulic machines** such as the hydraulic jack/lift used to raise vehicles in a garage, hydraulic brakes in cars, and the syringe.

Everyday Applications

Device	How pressure is applied
Drinking straw	Reduced pressure inside pulls liquid up
Syringe	Pushing the plunger increases pressure, forcing liquid out
Siphon	Uses atmospheric pressure to move liquid over a raised point
Hydraulic press/jack	Small force on small piston creates large force on large piston
Bicycle pump	Compresses air, increasing its pressure into the tyre

Key Inquiry: How does the density of a fluid, gravity, and depth affect the pressure inside it?

Quick Revision: $P = F/A$ for solids; $P = \rho gh$ for fluids. Pressure increases with depth and fluid density. Pascal's Principle explains hydraulic machines.

1.3 Mechanical Properties of Materials

Key Properties

Property	Meaning
Ductility	Ability to be drawn into a thin wire (e.g. copper)
Malleability	Ability to be hammered/rolled into sheets (e.g. aluminium)
Elasticity	Ability to return to original shape after a stretching/deforming force is removed
Brittleness	Breaks easily without bending (e.g. glass, chalk)
Strength	Ability to withstand force without breaking
Hardness	Resistance to scratching or denting
Stiffness	Resistance to bending/stretching

Hooke's Law

Hooke's Law states that the extension of a spring (or elastic material) is directly proportional to the applied force, provided the elastic limit is not exceeded.

$$F = ke \quad \text{where } F = \text{force (N)}, k = \text{spring/elastic constant (N/m)}, e = \text{extension (m)}$$

Stress, Strain and Young's Modulus

$$\text{Stress} = F/A \quad | \quad \text{Strain} = \Delta L / L_0 \quad | \quad \text{Young's Modulus (Y)} = \text{Stress} / \text{Strain}$$

where F = applied force, A = cross-sectional area, ΔL = extension, L_0 = original length.

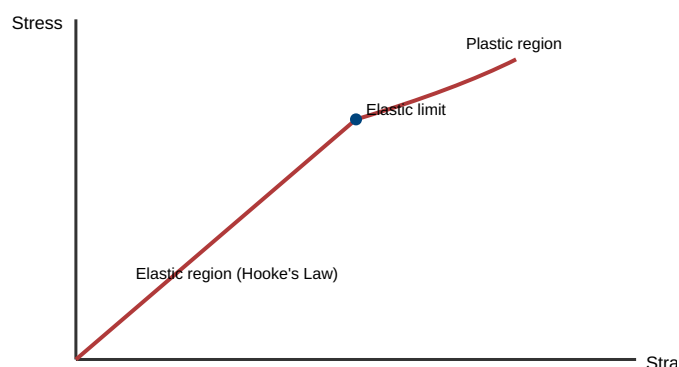


Fig 1.3: Stress–strain graph showing the elastic limit

Kenyan-context example: Steel reinforcement bars used in building construction (e.g. in Nairobi's high-rise buildings) are chosen for their high tensile strength and elasticity to withstand loads without snapping.

Quick Revision: Materials have properties like ductility, elasticity, and brittleness. Hooke's Law ($F = ke$) applies within the elastic limit. Stress/strain graphs reveal a material's elastic and plastic behaviour.

1.4 Temperature and Thermal Expansion

Temperature is a measure of the degree of hotness or coldness of a body, measured in degrees Celsius ($^{\circ}\text{C}$) or Kelvin (K).

Measuring Temperature

- **Liquid-in-glass thermometers** – use expansion of mercury/alcohol
- **Bimetallic strips** – two metals with different expansion rates bend when heated (used in thermostats)
- **Thermocouples** – generate voltage that changes with temperature
- **Thermistors/RTDs** – resistance changes with temperature
- **Infrared thermometers** – detect infrared radiation (used for body temperature checks)

Thermal Expansion

Most materials expand when heated and contract when cooled, because increased heat makes particles vibrate more and move further apart.



Fig 1.4: A bimetallic strip bends when heated because the two metals expand at different rates

Unusual Expansion of Water

Unlike most substances, water expands as it cools from 4°C to 0°C (and then freezes), which is why ice floats and why a full glass bottle of water can crack when frozen.

Applications of Thermal Expansion

- Gaps left between railway lines and bridge expansion joints to allow for expansion
- Thermostats in electric irons and fridges use bimetallic strips
- Sufuria (cooking pot) lids are made slightly wider to allow for expansion when heated
- Power lines are hung loosely to allow contraction on cold nights and expansion in heat

Key Inquiry: Why does a glass bottle full of water crack when the water inside freezes?

Quick Revision: Heat causes particles to move more and take up more space (expansion). Different materials expand at different rates — this is used in thermostats, bridges, and cooking equipment. Water is unusual: it expands when it freezes.

1.5 Moments and Equilibrium

The **moment of a force** (also called torque) is the turning effect of a force about a pivot (fulcrum).

Moment = Force × perpendicular distance from the pivot (Units: Newton-metres, Nm)

Principle of Moments

When a body is in equilibrium (balanced), the sum of clockwise moments equals the sum of anticlockwise moments about the same pivot.

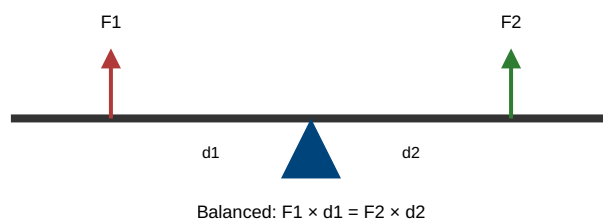


Fig 1.5: A see-saw (beam) balanced about a pivot — the principle of moments

Centre of Gravity (C.O.G) and Stability

The centre of gravity is the point where the whole weight of a body appears to act. A body is more stable when it has a **low centre of gravity** and a **wide base**.

State of Equilibrium	Description
Stable	Returns to original position when slightly displaced (e.g. a cone resting on its base)
Unstable	Falls further from original position when slightly displaced (e.g. a cone balanced on its tip)
Neutral	Stays in its new position when displaced (e.g. a ball on a flat surface)

Torque and Couple

A **couple** consists of two equal and opposite parallel forces that do not act along the same line, producing rotation only (no straight-line motion) — e.g. turning a steering wheel or a tap.

Kenyan-context example: Matatus and lorries are designed with a low centre of gravity and wide wheelbase to prevent them from toppling over on sharp bends — a key road safety consideration.

Quick Revision: Moment = Force $\times$ distance from pivot. Principle of moments: clockwise moments = anticlockwise moments at equilibrium. Stability depends on the position of the centre of gravity and base area.

1.6 Energy, Work, Power and Machines

Key Definitions

Work Done (W) = Force $\times$ distance moved in direction of force (Joules, J)
 Power (P) = Work Done $\div$ Time (Watts, W)

Forms of Mechanical Energy

$$\text{Kinetic Energy (K.E.)} = \frac{1}{2}mv^2 \quad | \quad \text{Gravitational Potential Energy (G.P.E.)} = mgh$$

The **Law of Conservation of Energy** states that energy cannot be created or destroyed, only transformed from one form to another.

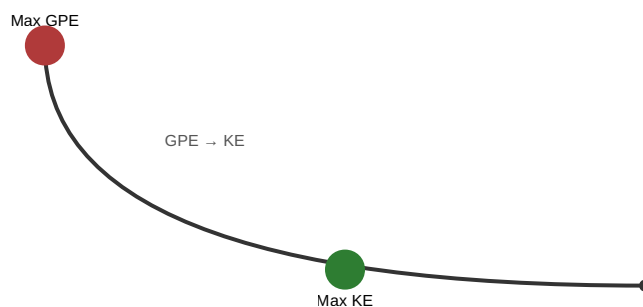


Fig 1.6: A ball rolling down a slope – gravitational potential energy converts to kinetic energy

Simple Machines

A machine is a device that makes work easier by changing the size or direction of an applied force.

$$\text{Mechanical Advantage (M.A.)} = \text{Load} \div \text{Effort}$$

$$\text{Velocity Ratio (V.R.)} = \text{Distance moved by effort} \div \text{Distance moved by load}$$

$$\text{Efficiency} = (\text{M.A.} \div \text{V.R.}) \times 100\%$$

Machine	Example / Use
Lever	Wheelbarrow, jembe (hoe), crowbar
Inclined plane	Ramp for loading a lorry
Pulley	Flag hoist, well/borehole water bucket
Wheel and axle	Steering wheel, bicycle pedal
Gears	Bicycle chain, motorbike engine
Screw	Bottle cap, vice, jack
Hydraulic lift	Car garage vehicle lift

Modern applications: Elevators (lifts), escalators in shopping malls, and excavators used in construction all combine several simple machines to make heavy work easier and faster.

Key Inquiry: How do machines make work easier, and why is no machine 100% efficient in practice?

Quick Revision: Work = Force $\times$ distance; Power = Work $\div$ time. Energy is conserved but changes form (e.g. GPE $\leftrightarrow$ KE). Machines change force size/direction; efficiency = (MA/VR) $\times$ 100%.

STRAND 2: Waves and Optics

How waves travel, behave, and how unstable atoms release energy through radioactivity

2.1 Properties of Waves

A **wave** is a disturbance that transfers energy from one point to another without transferring matter.

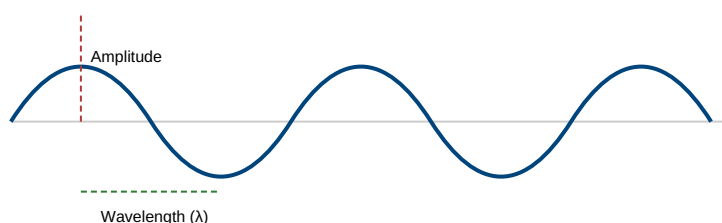


Fig 2.1: Parts of a transverse wave

$$\text{Wave speed (v)} = \text{frequency (f)} \times \text{wavelength (\lambda)}$$

Key Wave Properties (Qualitative)

Property	Meaning	Everyday example
Rectilinear propagation	Waves travel in straight lines	Light forming sharp shadows
Reflection	A wave bounces back after hitting a surface	Echo off a hillside; mirror image
Refraction	A wave changes speed and direction when moving into another medium	A straw appears bent in a glass of water
Diffraction	A wave spreads out after passing through a gap or around an obstacle	Hearing sound around a corner
Interference	Two waves combine (add up or cancel out)	Sound "beats" from two speakers

Stationary (Standing) Waves and Resonance

A stationary wave forms when two identical waves travelling in opposite directions overlap, producing fixed points of no vibration (**nodes**) and points of maximum vibration (**antinodes**). This occurs in vibrating strings (e.g. a guitar string) and air columns (e.g. a flute or organ pipe). **Resonance** occurs when a vibrating system is driven at its natural frequency, producing a large amplitude of vibration.

Doppler Effect

The Doppler effect is the change in observed frequency (pitch) of a wave when the source and observer are moving relative to each other.

Everyday example: The siren of an approaching ambulance sounds higher-pitched, then suddenly drops to a lower pitch as it passes and moves away.

Key Inquiry: Where is the Doppler effect applied in real-life situations, such as speed radar guns and medical ultrasound?

Quick Revision: $v = f\lambda$. Waves show reflection, refraction, diffraction and interference. Stationary waves have nodes and antinodes. The Doppler effect explains why pitch changes as a source moves past you.

2.2 Radioactivity and Stability of Isotopes

Radioactivity is the spontaneous emission of radiation (particles or energy) from an unstable atomic nucleus as it changes into a more stable form.

Types of Radioactive Emissions

Radiation	Nature	Charge	Penetration Power
Alpha (α)	Helium nucleus (2 protons + 2 neutrons)	+2	Stopped by paper/skin
Beta (β)	Fast-moving electron	-1	Stopped by a few mm of aluminium
Gamma (γ)	High-energy electromagnetic wave	0	Needs thick lead or concrete to stop

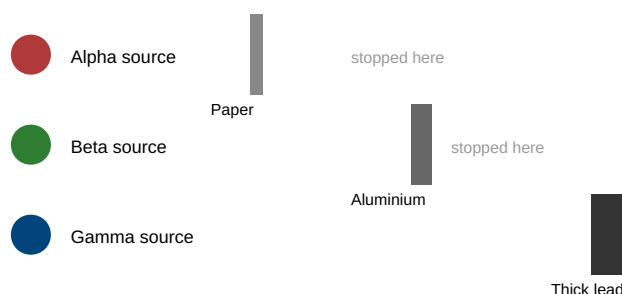


Fig 2.2: Penetrating power of alpha, beta and gamma radiation

Nuclear Stability and Decay

An unstable nucleus (radionuclide) emits radiation to become more stable, changing into a different element in the process (radioactive decay). This is represented using nuclear equations, balancing mass number and atomic number.

Half-Life

The **half-life** ($t_{1/2}$) of a radioactive element is the time taken for half of the radioactive nuclei in a sample to decay.

$$N = N_0 (1/2)^{T/t_{1/2}}$$

where N_0 = initial number of atoms, N = number remaining, T = total time elapsed

Nuclear Fission and Fusion

- **Nuclear fission** – a large nucleus splits into smaller nuclei, releasing energy (used in nuclear power plants)
- **Nuclear fusion** – small nuclei combine to form a larger nucleus, releasing enormous energy (this is how the sun produces its energy)

Applications and Safety

Field	Application
Medicine	Cancer treatment (radiotherapy), sterilising equipment
Agriculture	Pest control, improving crop varieties
Industry	Checking for cracks in metal pipes/welds
Archaeology	Carbon dating to determine the age of fossils

Safety note: Radioactive materials must be handled with lead shielding, limited exposure time, and safe disposal to avoid harm — as highlighted by disasters like Chernobyl.

Key Inquiry: What are the risks of exposure to radiation, and how can these risks be minimised?

Quick Revision: Alpha, beta and gamma radiations differ in penetration power. Half-life is the time for half a sample to decay. Fission splits nuclei; fusion joins them. Radioactivity has valuable uses but needs strict safety precautions.

STRAND 3: Electricity and Magnetism

Static and current electricity, and an introduction to electronic materials

3.1 Electrostatics

Static electricity is caused by the transfer of electrons between materials, leaving one object positively charged (fewer electrons) and the other negatively charged (more electrons).

Charging Methods

Method	Description
By friction (rubbing)	Rubbing two insulators transfers electrons, e.g. a plastic ruler rubbed on hair attracts paper bits
By contact	A charged object touches a neutral conductor, sharing its charge
By induction	A charged object is brought near (not touching) a conductor, separating its charges

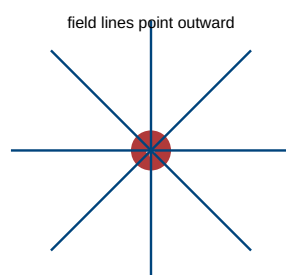


Fig 3.1: Electric field lines around a positive point charge, pointing outward

The Gold-Leaf Electroscope

An electroscope detects and tests the type and amount of electric charge on an object using a thin metal (gold or foil) leaf that diverges when charged.

Applications of Electrostatics

- **Spray guns and paint sprayers** – charged paint droplets spread evenly and stick to a surface
- **Photocopiers and laser printers** – use static charge to attract toner/ink to paper
- **Electrostatic precipitators** – remove dust/smoke particles from factory chimneys
- **Lightning** – a natural, large-scale discharge of static electricity between clouds and the ground; lightning arrestors on tall buildings provide a safe path for this charge to reach the ground

Key Inquiry: How do lightning arrestors protect tall buildings from lightning strikes?

Quick Revision: Charges are transferred by friction, contact or induction. Like charges repel; unlike charges attract. An electroscope detects charge. Static electricity is applied in spray painting, photocopying, and lightning protection.

3.2 Current Electricity

Key Terms

Term	Meaning	Unit
Current (I)	Rate of flow of charge	Ampere (A)
Potential difference (V)	Energy transferred per unit charge between two points	Volt (V)
Resistance (R)	Opposition to the flow of current	Ohm (Ω)
E.M.F. (E)	Total energy supplied per unit charge by a source (cell/battery)	Volt (V)

Ohm's Law

Ohm's Law states that the current through a conductor is directly proportional to the potential difference across it, provided temperature remains constant.

$$V = IR$$

$$E = I(R + r) \quad (r = \text{internal resistance of the cell})$$

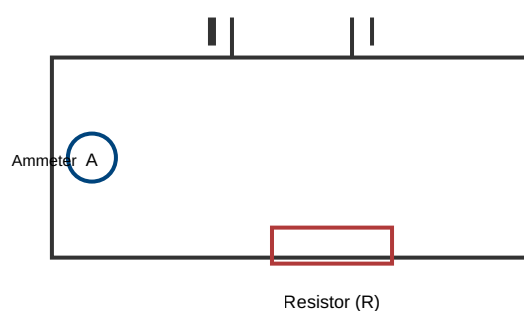


Fig 3.2: A simple series circuit with a cell, ammeter and resistor

Resistors

Resistors control current flow. Fixed resistors are often colour-coded to show their resistance value. Resistors can be connected:

- **In series** – total resistance $R = R_1 + R_2 + R_3 \dots$ (current is the same throughout)
- **In parallel** – $1/R = 1/R_1 + 1/R_2 + 1/R_3 \dots$ (voltage is the same across each branch)

Electrical Heating and Power

$$\text{Power (P)} = VI = I^2R = V^2/R$$

Everyday application: Electric jikos, kettles, and irons use the heating effect of current. Understanding power ratings (in watts) helps households manage electricity (KPLC/Kenya Power) bills and choose the right fuse for safety.

Quick Revision: $V = IR$ (Ohm's Law); $E = I(R+r)$ for a full circuit. Resistors can be arranged in series or parallel. Electrical power $P = VI$ is the basis of electric heating devices.

3.3 Introduction to Electronics

Classifying Materials by Conductivity

Material Type	Description	Example
Conductor	Allows electric current to flow easily (many free electrons)	Copper, silver, aluminium
Insulator	Does not allow current to flow (electrons tightly bound)	Rubber, glass, plastic
Semiconductor	Conducts moderately; conductivity increases with temperature	Silicon, germanium
Superconductor	Has zero electrical resistance below a very low critical temperature	Certain metals/alloys at extremely low temperatures

Semiconductors

- **Intrinsic semiconductors** – pure semiconductor material (e.g. pure silicon)
- **Extrinsic semiconductors** – formed by adding small amounts of impurity (doping) to improve conductivity:
 - **n-type** – doped to have extra free electrons
 - **p-type** – doped to have "holes" (missing electrons) that act as positive charge carriers

Applications

Real-world uses: Semiconductors are the basis of diodes, transistors, and computer chips found in phones, radios, solar panels, and solar-powered lanterns widely used in rural Kenya.

Key Inquiry: How does temperature affect the resistance of conductors compared to semiconductors?

Quick Revision: Materials are classified as conductors, insulators, semiconductors, or superconductors based on how easily they conduct current. Semiconductors can be doped into n-type or p-type, forming the basis of modern electronics.

STRAND 4: Environmental and Space Physics

Understanding our planet's climate system and our place in the universe

4.1 Greenhouse Effect and Climate Change

The **greenhouse effect** is the natural process by which certain gases in the atmosphere (greenhouse gases) trap heat from the sun, keeping the Earth warm enough to support life.

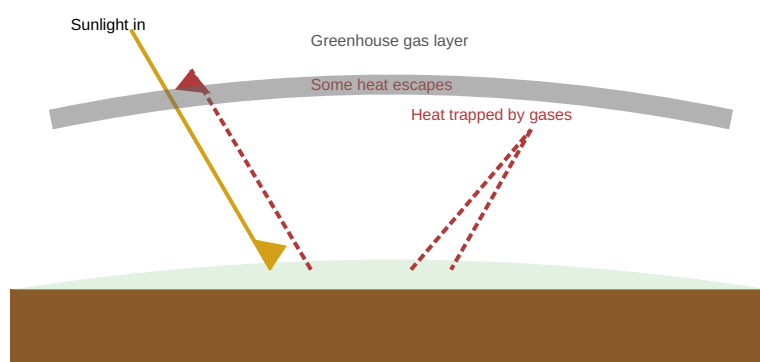


Fig 4.1: The greenhouse effect – gases trap some of the sun's heat near the Earth's surface

Greenhouse Gases

Carbon dioxide (CO_2), methane (CH_4), and water vapour are the main greenhouse gases. Human activities — burning fossil fuels, deforestation, and industrial processes — increase the concentration of these gases, intensifying the greenhouse effect (**global warming**).

The Ozone Layer

The ozone layer (in the upper atmosphere) absorbs most of the sun's harmful ultraviolet (UV) radiation. Depletion of the ozone layer (caused by chemicals like CFCs) allows more UV radiation to reach the Earth's surface, increasing risks such as skin cancer.

Effects of Climate Change

- Melting glaciers and rising sea levels
- Changing rainfall patterns – more frequent droughts in areas like northern Kenya, and flooding in others
- Shrinking water bodies (e.g. changing water levels in Lake Victoria and other lakes)
- Shifts in growing seasons affecting agriculture

Mitigating Climate Change

Kenyan initiatives: Tree-planting campaigns, use of renewable energy (geothermal at Olkaria, solar, wind at Lake Turkana Wind Farm), and the ban on single-use plastic bags all help reduce environmental impact.

Key Inquiry: How do human actions impact climate change, and how does ozone layer depletion threaten our environment?

Quick Revision: Greenhouse gases trap heat, warming the Earth. Human activity increases this effect (global warming). The ozone layer protects us from UV radiation. Renewable energy and conservation help mitigate climate change.

4.2 Introduction to Space Physics

Origin of the Universe

The **Big Bang Theory** proposes that the universe began as an extremely hot, dense point about 13.8 billion years ago, and has been expanding and cooling ever since.

Classification of Celestial Bodies

Body	Description
Star	A huge ball of hot gas that produces its own light and heat through nuclear fusion (e.g. the Sun)
Planet	A large body that orbits a star and does not produce its own light (e.g. Earth, Mars)
Moon	A natural satellite that orbits a planet
Comet	An icy body that develops a glowing tail as it nears the sun
Asteroid	A small rocky body, smaller than a planet, mostly found in the asteroid belt

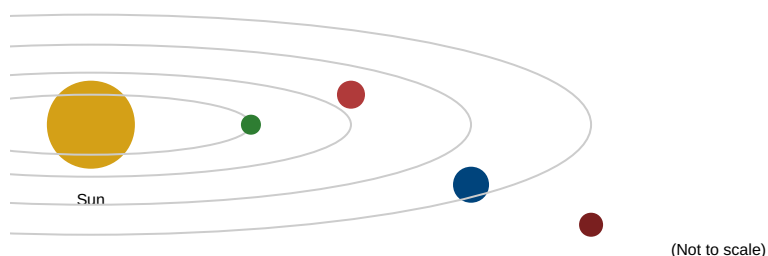


Fig 4.2: Simplified model of planets orbiting the sun

Planetary Motion

Planets orbit the sun in elliptical (oval-shaped) paths due to the sun's gravitational pull. This is described by Kepler's laws of planetary motion.

Telescopy and Space Exploration

Telescopes collect and magnify light from distant objects, allowing astronomers to study stars, planets, and galaxies. Space exploration has advanced from early telescopes to satellites, space probes, and crewed missions (e.g. the Apollo Moon landings, the International Space Station).

Careers in Space Science

Astronomer

Astrophysicist

Aerospace Engineer

Satellite Technician

Astronaut

Remote Sensing Analyst

Did you know? Kenya has its own space agency — the Kenya Space Agency (KSA) — which coordinates satellite and space-related activities in the country.

Key Inquiry: How was the universe/Earth formed, and how do we benefit from astrophysics?

Quick Revision: The Big Bang Theory explains the universe's origin. Celestial bodies include stars, planets, moons, comets and asteroids. Planets orbit the sun in elliptical paths. Telescopes and space exploration help us study and benefit from space.